Experiment 02

Learning and Assignment of IP Address manually to computers

Aim

To learn IPv4 addressing and manually assign unique static addresses to two lab computers.

Requirements

Two lab computers, switch, Ethernet cables; Windows IPv4 settings or Linux NetworkManager.

Procedure / algorithm

1. Identify the assigned Ethernet interface and record the current DHCP configuration.

2. Connect both PCs to the switch; use the addressing table and check that the chosen addresses are unused.

3. Set the IPv4 address, /24 mask and gateway on each PC. Configure DNS only when the specified service exists.

4. Verify the address and connected route; ping the peer and gateway. Check the firewall ICMP setting if replies are blocked.

5. Compare the network portion 192.168.20 with the distinct host portions 11 and 12; restore the original configuration after the lab.

Complete source / configuration: addressing.csv

device	IPv4	mask	gateway	DNS
PC1	192.168.20.11	255.255.255.0	192.168.20.1	192.168.20.1
PC2	192.168.20.12	255.255.255.0	192.168.20.1	192.168.20.1
Router	192.168.20.1	255.255.255.0	none	none

Complete source / configuration: configuration.txt

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Windows graphical configuration:
Settings > Network & Internet > Advanced network settings > More network adapter options.
Open the lab Ethernet adapter Properties > Internet Protocol Version 4 (TCP/IPv4).
Select Use the following IP address; enter the assigned row in addressing.csv.
Enter DNS only if 192.168.20.1 provides the lab DNS service.
Verify: ipconfig /all
Verify PC1 to PC2: ping 192.168.20.12
Verify gateway: ping 192.168.20.1
Restore Obtain an IP address automatically / DNS automatically after the lab.

Linux NetworkManager (replace LAB-ETH with the lab connection name):
nmcli connection show
sudo nmcli connection modify 'LAB-ETH' ipv4.method manual ipv4.addresses 192.168.20.11/24
    ipv4.gateway 192.168.20.1 ipv4.dns 192.168.20.1
sudo nmcli connection up 'LAB-ETH'
ip -4 address show
ip route show
ping -c 4 192.168.20.12
Restore: sudo nmcli connection modify 'LAB-ETH' ipv4.method auto ipv4.addresses '' ipv4.gateway ''
    ipv4.dns ''
sudo nmcli connection up 'LAB-ETH'
```

Actual output / verification

Not executed in the required lab/tool environment. No actual operational output is claimed.

Expected output - not an execution record

Both PCs show their assigned addresses and can exchange ICMP replies; a configured, connected gateway replies.

Observations

A /24 mask places both addresses in the same subnet. A gateway is needed for off-subnet traffic; DNS resolves names, not numeric addresses.

Result

The static-address plan and complete configuration steps are supplied. Live interface settings were not changed here.